

Duality

Duality in the discrete FS

Consider $f[m] = \frac{1}{N} \sum_{r=\langle N \rangle} g[r] e^{-jr(2\pi/N)m}$ f[m] and g[r] are periodic

$m = k, r = n$ $\Rightarrow$ $m = n, r = -k$

$$f[k] = \frac{1}{N} \sum_{n=\langle N \rangle} g[n] e^{-jk(2\pi/N)n}$$

$$f[n] = \frac{1}{N} \sum_{k=\langle N \rangle} g[-k] e^{jk(2\pi/N)n}$$

$$g[n] \xrightarrow{\mathcal{FS}} f[k]$$

$$f[n] \xrightarrow{\mathcal{FS}} \frac{1}{N} g[-k]$$

Every property of the discrete FS has a dual.

Examples

$$\left\{ \begin{array}{l} x[n - n_0] \xleftrightarrow{\mathcal{FS}} a_k e^{-jk(2\pi/N)n_0} \\ e^{jm(2\pi/N)n} x[n] \xleftrightarrow{\mathcal{FS}} a_{k-m} \end{array} \right. \quad \left\{ \begin{array}{l} \sum_{r=\langle N \rangle} x[r] y[n - r] \xleftrightarrow{\mathcal{FS}} N a_k b_k \\ x[n] y[n] \xleftrightarrow{\mathcal{FS}} \sum_{l=\langle N \rangle} a_l b_{k-l} \end{array} \right.$$

Handwritten notes:

$$\begin{array}{ccc} f[n] & & \frac{1}{N} g[-k] \\ & \swarrow \searrow & \\ g[n] & & f[k] \end{array}$$

所有周期都一样

Duality



Duality in the discrete FS: examples

$$x[n] = \begin{cases} \frac{1}{9} \frac{\sin(5\pi n/9)}{\sin(\pi n/9)}, & n \neq \text{multiple of } 9 \\ 5/9, & n = \text{multiple of } 9 \end{cases}$$

$$a_k = ?$$

Duality in the discrete FS

① Dual in the frequency domain: $n \rightarrow k$

② Time domain signal

$$b_k = \begin{cases} \frac{1}{9} \frac{\sin(5\pi k/9)}{\sin(\pi k/9)}, & k \neq \text{multiple of } 9 \\ 5/9, & k = \text{multiple of } 9 \end{cases}$$

$g[n] \xleftrightarrow{\mathcal{FS}} b_k$

Duality



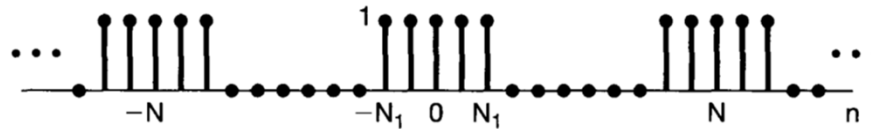
Duality in the discrete FS: examples

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$$a_k = ?$$

$N_1 = 2, N = 9$

Recall



$\updownarrow \mathcal{FS}$

$$\begin{cases} \frac{1}{N} \frac{\sin[2\pi k(N_1 + 1/2)/N]}{\sin(\pi k/N)}, & k \neq 0, \pm N, \pm 2N, \dots \\ (2N_1 + 1)/N, & k = 0, \pm N, \pm 2N, \dots \end{cases}$$

Duality



Duality in the discrete FS: examples

$$x[n] = \begin{cases} \frac{1}{9} \frac{\sin(5\pi n/9)}{\sin(\pi n/9)}, & n \neq \text{multiple of } 9 \\ 5/9, & n = \text{multiple of } 9 \end{cases}$$

$$a_k = \frac{1}{N} g[-k] = \begin{cases} 1/9, & |k| \leq 2 \\ 0, & 2 < |k| \leq 4 \end{cases}$$

③ Duality

$$g[n] = \begin{cases} 1, & |n| \leq 2 \\ 0, & 2 < |n| \leq 4 \end{cases}$$

② Time domain signal

① Dual in the frequency domain: $n \rightarrow k$

$g[n] \xleftrightarrow{\mathcal{FS}} b_k$

$$b_k = \begin{cases} \frac{1}{9} \frac{\sin(5\pi k/9)}{\sin(\pi k/9)}, & k \neq \text{multiple of } 9 \\ 5/9, & k = \text{multiple of } 9 \end{cases}$$

$g[n]$ is periodic ($N = 9$)

Duality



Duality between discrete FT and continuous FS

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega \quad \text{Discrete FT} \quad X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n] e^{-j\omega n}$$

Non-periodic, discrete in time

Periodic, continuous in frequency

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$$

Periodic, continuous in time

Continuous FS

$$a_k = \frac{1}{T} \int_T x(t) e^{-jk\omega_0 t} dt$$

Non-periodic, discrete in frequency

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jkt}$$

$\omega_0 = 1$

$$a_k = \frac{1}{T} \int_T x(t) e^{-jkt} dt$$



Properties of discrete-time Fourier Transform

Duality between discrete FT and continuous FS: Examples

$$x[n] = \frac{\sin(\pi n/2)}{\pi n}$$

$$X(e^{j\omega}) = ?$$

① DT $\rightarrow$ CT FS: $n \rightarrow k$

$$g(t) = \begin{cases} 1, & |t| \leq 0.5\pi \\ 0, & 0.5\pi < |t| \leq \pi \end{cases}$$

FS

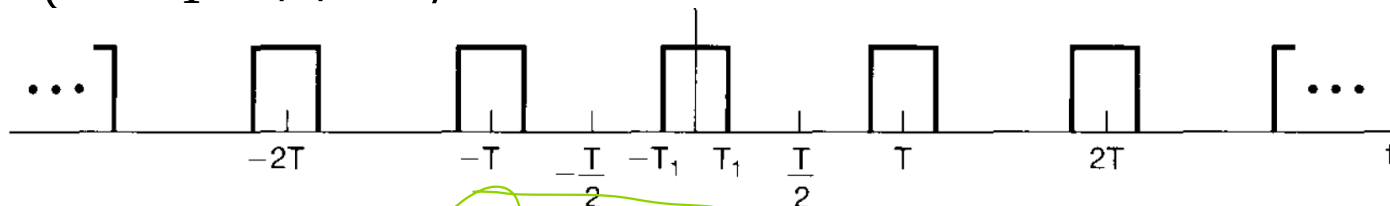
② CT signal

$$a_k = \frac{\sin(k\pi/2)}{k\pi}$$

$$g(t) = \begin{cases} 1, & |t| \leq T_1 \\ 0, & T_1 < |t| \leq T/2 \end{cases}$$

FS

$$a_k = \frac{\sin(k\omega_0 T_1)}{k\pi}, k \neq 0$$



$$T = 2\pi, \omega_0 = 1, T_1 = 0.5\pi$$



Properties of discrete-time Fourier Transform

Duality between discrete FT and continuous FS: Examples

$$\begin{aligned}
 x[n] &= \frac{\sin(\pi n/2)}{\pi n} & X(e^{j\omega}) &=? \\
 g(t) &= \begin{cases} 1, & |t| \leq \pi/2 \\ 0, & \pi/2 < |t| \leq \pi \end{cases} & a_k &= \frac{\sin(k\pi/2)}{k\pi}
 \end{aligned}$$

① $n \rightarrow k$

② CT signal

$$\frac{\sin(\pi k/2)}{\pi k} = \frac{1}{T} \int_T g(t) e^{-jk\omega_0 t} dt = \frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} g(t) e^{-jkt} dt \quad (\omega_0 = 1)$$

$n = k, t = \omega$

$$\frac{\sin(\pi n/2)}{\pi n} = \frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} g(\omega) e^{-j\omega n} d\omega \quad \longrightarrow \quad \frac{\sin(\pi n/2)}{\pi n} = \frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} g(-\omega) e^{j\omega n} d\omega$$

$$\therefore X(e^{j\omega}) = g(-\omega) = \begin{cases} 1, & |\omega| \leq \pi/2 \\ 0, & \pi/2 < |\omega| \leq \pi \end{cases}$$